

Microplastic Polymer Analysis Report

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Sample ID	FTIR-8028

Prediction Results

Detected Polymer	HDPE
Confidence Score	0.94%
Model Used	finetuned_model

FTIR Spectral Analysis

- 2915 cm⁻¹ → CH₂ asymmetric stretching
- 1470 cm⁻¹ → CH₂ bending

Confidence Interpretation

■■■■■■■■■■ 0.94%

Scientific References

[1] Thompson, R. C., Olsen, Y., Mitchell, R. P., Davis, A., Rowland, S. J., John, A. W. G., McGonigle, D., & Russell, A. E. (2004). Lost at sea: Where is all the plastic? Science, 304(5672), 838. DOI: <https://doi.org/10.1126/science.1094559>

[2] Barnes, D. K. A., Galgani, F., Thompson, R. C., & Barlaz, M. (2009). Accumulation and fragmentation of plastic debris in global environments. Philosophical Transactions of the Royal Society B, 364(1526), 1985–1998. DOI: <https://doi.org/10.1098/rstb.2008.0205>

[3] Gewert, B., Plassmann, M. M., & MacLeod, M. (2015). Pathways for degradation of plastic polymers in marine environments. Environmental Science: Processes & Impacts, 17(9), 1513–1521. DOI: <https://doi.org/10.1039/C5EM00207A>

[4] Jung, M. R., et al. (2018). Validation of ATR FT-IR to identify polymers of plastic marine debris. Marine Pollution Bulletin, 127, 704–716. DOI: <https://doi.org/10.1016/j.marpolbul.2017.12.061>

[5] Pimpke, S., et al. (2018). Reference database design for FTIR-based automated microplastic analysis. Analytical and Bioanalytical Chemistry, 410(21), 5131–5141. DOI: <https://doi.org/10.1007/s00216-018-1156-x>

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